

Total No. of Questions : 10]

SEAT No. :

PD-5305

[Total No. of Pages : 2

[6403]-310R

T.E. (Honours in Artificial Intelligence and Machine Learning)

COMPUTATIONAL STATISTICS

(2019 Pattern) (Semester-V) (310301)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Make suitable assumptions whenever necessary.

Q1) a) What is Mean, Median and Mode with example? [5]

b) Difference between Univariate and Bivariate Analysis? [5]

OR

Q2) a) Explain the Histogram, Statistical Thinking with example? [5]

b) Difference between the Standard Deviation & Harmonic Mean with example? [5]

Q3) a) Explain Normal Distributions with example? [5]

b) Difference between the Characterizing a Distribution & Discrete Distributions? [5]

OR

Q4) a) Write down short note on Lognormal, Weigh bull, Exponential, Uniform? [5]

b) What is Poisson Distribution with suitable example? [5]

P.T.O.

- Q5) a)** Explain the Hypothesis Concept with example. [8]
b) What is Confusion Matrix in detail. [8]

OR

- Q6) a)** What is Test on Numerical Data- Distribution of a Sample Mean with example. [8]
b) Comparison of Two Groups Multiple Groups. [8]
- Q7) a)** Write down the short no Normalization- Feature Scaling. [8]
b) Difference between the Ridge Regression, Lasso Regression. [8]

OR

- Q8) a)** Explain in detail Cross Validation Techniques- K-fold with example. [8]
b) Difference between the Grid Search CV & CV Error. [8]
- Q9) a)** What is the different Dimensionality Reduction Techniques. Explain with example. [9]
b) Where we use the Chi2 square method with example. [9]

OR

- Q10) a)** When we used the Correlation coefficient, Rank Correlation & how with example. [9]
b) Difference between the Bayes' Theorem & Monte Carlo Method [9]

